PREDICTIVE MAINTENANCE IN METRO TRAINS USING MACHINE LEARNING AND DEEP LEARNING TECHNIQUES

A Dual-Evaluation Benchmark of LSTM, GRU and TCN on the MetroPT-3 and MetroPT-2 Compressor Datasets

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Source code: https://github.com/Chetan0246/MetroTrain-Predictive-Maintenance

Abstract

Project. This project builds and evaluates an end-to-end predictive-maintenance pipeline for the Air Production Unit (APU) compressor of Metro do Porto passenger trains. It benchmarks 10 deep learning architectures spanning four sequential paradigms (recurrent, hybrid convolutional, multi-head transformer, and unsupervised generative models) on real-world multi-gigabyte industrial telemetry datasets (MetroPT-2 and MetroPT-3) under strict zero-leakage chronological partitions.

Base paper. Jakobs, Veloso and Gama (2026) [1] learn short, human-readable rules for online failure prediction on MetroPT-2. Their method uses online rule learning (RIPPER, Decision Trees) and tree ensembles, reporting initial failure coverage but suffering high false alarm rates under severe class imbalance and sensor drift. They identify out-of-distribution evaluation on MetroPT-3 and architectures without selective sensors as key open challenges.

Proposed method. We evaluate 10 deep sequence models across 10-minute windows of 16 APU channels. On MetroPT-2 (23,668 held-out test sequences), Variational Autoencoder (VAE) and LSTM Autoencoder achieve peak PR-AUC of **0.9833** (100% recall); Temporal Fusion Transformer (TFT) delivers the top operational deployment profile with **0.9817 PR-AUC**, **99.95% recall**, **96.74% precision**, and **0.9832 F1-score**; GRU achieves top F1 of **0.9842** (PR-AUC 0.9560); while CNN-BiLSTM with Attention delivers 100.00% recall with 0.05 ms latency. On MetroPT-3 (26,925 held-out sequences with unseen lubrication and intake valve failures), Transformer Encoder leads with **0.1714 PR-AUC**, followed by BiLSTM (0.1398), TFT (0.1381), and TCN (0.1378). Unsupervised validation-quantile threshold calibration (q_{90}) successfully recovers recall from 0.00% collapse to 35.02% without test label leakage.

New features. (1) A comprehensive 10-algorithm deep learning benchmark suite across 4 sequential paradigms. (2) Dual-Evaluation framework isolating active fault state monitoring (Task 1) from 2-hour anticipatory pre-failure warning (Task 2, active outage excluded). (3) Strict zero-leakage chronological partitioning (65/15/20 and 70/15/15). (4) 1:1 balanced temporal sliding window undersampling evaluated against SMOTE. (5) Unsupervised validation-only quantile threshold calibration (q_{90}, q_{95}, q_{98}) resolving out-of-distribution recall collapse without test label exposure. (6) Sub-millisecond edge latency benchmarking (<0.5 ms per sequence).

1. Introduction

Metro trains carry large numbers of passengers every day, so their reliability depends on mechanical, electrical, pneumatic and control subsystems working together. When one of them fails, the result is a delayed service, an emergency inspection or an unplanned withdrawal from service. Maintenance has traditionally been reactive (repair after a fault) or time-based (service at fixed intervals). Reactive maintenance causes unplanned downtime, and fixed-interval maintenance can replace parts that still have life left. Predictive maintenance uses the measured condition of the equipment as evidence for when to act [27], [29].

Metro systems suit data-driven maintenance because every train carries many continuously monitored subsystems. The Air Production Unit (APU) is a good example. Its compressor supplies compressed air to the pneumatic brakes, the secondary suspension, the pantograph and the passenger doors, so a compressor

failure is a safety and service problem. Pressure, temperature, motor current, flow and valve states can be read together as a multivariate time series, and a model can learn the difference between normal and abnormal compressor behaviour [28].

This project makes the following contributions:

1. **Dual-evaluation benchmark.** Task 1 (active fault detection, warning window plus outage) and Task 2 (anticipatory warning only, outage excluded) are scored separately, so detecting a breakdown that has already happened is never counted as early warning.
2. **Leak-free pipeline.** Strict chronological splits (MetroPT-3 70/15/15, MetroPT-2 65/15/20), scaler fitted on training data only, windowing inside each split, and 1:1 balancing applied to the training set only. Validation and test sets keep their natural imbalance.
3. **Six reproducible experiments.** LSTM, GRU and TCN on both datasets, with fixed seeds and the full 10.77M and 7.12M raw records processed in streaming chunks.
4. **Validation-only threshold calibration.** Operating thresholds from unsupervised quantiles (q90, q95, q98) of healthy validation scores fix the recall collapse of the default 0.5 threshold without touching test labels.
5. **Base-rate and shift analysis.** Closed-form explanation of the ~49.6% warning precision, and a measured cross-fault generalisation gap (air leak to oil leak) with its seasonal false-positive drift.

2. Literature Survey

Fifteen recent papers on predictive maintenance in metro, railway and closely related rotating-machinery settings were reviewed in Review 1. They fall into four groups: metro and APU benchmarks and rule-based prediction [1], [3]; bearing and gearbox fault diagnosis [2], [4], [8], [9], [10], [15]; bogie, wheel and wheelset monitoring [5], [6], [11]; and railway track and multimodal inspection [7], [12], [13], [14].

(a) Theory

Predictive maintenance as time-series classification. Each sample is a window of multivariate sensor readings, and the label says whether the window lies in a documented pre-failure or failure interval. Classical models (Random Forest, XGBoost) work on engineered window features. Deep sequence models learn the temporal structure directly from the raw window [28].

LSTM. A Long Short-Term Memory network keeps a cell state that is protected by input, forget and output gates, which lets it carry information across many time steps without the vanishing-gradient problem of plain RNNs [17]. **GRU.** A Gated Recurrent Unit merges the gates into a reset gate and an update gate and has no separate cell state, so it has fewer parameters and trains faster [18]. **TCN.** A Temporal Convolutional Network stacks dilated causal convolutions with residual connections. Dilation grows the receptive field exponentially with depth while causality guarantees that no future step is used, and the whole sequence is processed in parallel [19]. A TCN-LSTM hybrid has also been used for bearing remaining-useful-life prediction [10].

Imbalance and metrics. Under extreme imbalance, the Precision-Recall curve and its area (PR-AUC) are more informative than ROC-AUC or accuracy, because precision responds directly to false alarms against a very large healthy class [20], [21]. Even a small false-positive rate produces many false alarms when negatives outnumber positives by 166 to 1, which is the base-rate effect analysed in Section 5.

Leak-free evaluation and thresholds. For time series, random cross-validation lets training windows overlap the test period, so evaluation must respect chronology [25]. A classifier trained on a 1:1 balanced set outputs scores that are not calibrated to the natural prevalence, so the default 0.5 threshold can be badly wrong at test time. Choosing the threshold from healthy validation scores is a label-free remedy [26].

Interpretable online prediction. Jakobs et al. [1] learn short rules on MetroPT-2 that human operators can read. This supports using a deep model as an alarm generator and an interpretable layer as the explanation, which the present project lists as future work.

(b) Comparison of reviewed papers

Table 1. *Summary of the 15 papers reviewed in Review 1.*

#	Authors (year)	Data	Method	Best reported result / contribution
[1]	Jakobs et al. (2026)	MetroPT2	Online rule learning + failure prediction	High failure coverage with short interpretable rules
[2]	Neupane et al. (2026)	Paderborn bearing data	Scalogram/kurtogram fusion; IF, OC-SVM, AE, VAE	Several sensor combinations with F1/AUC above 0.95
[3]	Steiner et al. (2026)	MetroAT (105 variables, 25M points)	Data-selection pipeline + K-means	New open metro pneumatic benchmark covering APU, brake and levelling
[4]	Sun et al. (2026)	SEU dataset	CWT + CNN + BKA + LSSVM	Beats comparison models on small samples
[5]	Wang et al. (2026)	BJTU-RAO bogie	Selective knowledge distillation + domain adaptation	~2.1% cross-domain gain; model about 28.5 kB
[6]	Shaikh et al. (2025)	FaultSeg wheel images	YOLOv9 instance segmentation	~87% accuracy on validation task
[7]	Ma et al. (2026)	Rail-VIVID	Synchronised vibration + vision benchmark	First controlled multimodal track-anomaly benchmark
[8]	Sehri et al. (2025)	UORED-VAFCLS	CNN + LSTM, vibration-acoustic fusion	Cross-domain bearing diagnosis
[9]	Yu et al. (2025)	MFPT	CWT + GAN + multi-size kernel attention	Better small-sample diagnosis
[10]	Liu et al. (2025)	XJTU-SY bearings	TCN + LSTM + CBAM	MAE 2.287; RMSE 3.123 (RUL)
[11]	Song et al. (2024)	WLI-Set	Laser-image segmentation, centreline extraction	Public wheelset image benchmark
[12]	Arain et al. (2024)	Railway track surface faults	Image-based fault recognition	Seven defect classes from inspection vehicle
[13]	Abisheva et al. (2024)	RailVista	Dataset construction and annotation	200,000 images, 19 defect classes
[14]	Eunus et al. (2024)	Railway track faults	Conv. AE + ResNet + LSTM ensemble	93.28% accuracy
[15]	Huang (2024)	JNU bearings	Parallel residual net + MK-MMD	~97.79% average accuracy (six transfer tasks)

(c) Limitations of existing work and major contribution of the proposed work

Limitations of existing work

1. Few metro-pneumatic studies with deep temporal models. Only [1] and [3] work on metro APU or pneumatic data. [1] is rule-based and evaluated on MetroPT-2, and lists MetroPT-3 as future work. This project benchmarks deep sequence models on both datasets.
2. Curated, small-sample or image data. Most reviewed studies use bearing test-rig or image datasets [2], [4], [8], [9], [12]–[15]. They do not face the 96–99% healthy-class imbalance of real fleet telemetry. This project keeps natural imbalance in validation and test sets.
3. Detection and warning are not separated. Reported scores rarely distinguish detecting an ongoing fault from warning before it starts. Task 1 and Task 2 here are scored independently, with the active outage excluded from Task 2.
4. Generalisation across failure modes is rarely tested. Training on one fault family and testing on another is uncommon in the reviewed work. Here, the MetroPT-3 test failure is an oil leak that never appears in training.
5. Thresholds are taken for granted. A default 0.5 cut-off on a balanced-trained model is common. This project shows it collapses recall to zero and fixes it with validation-only calibration.

Major contribution of the proposed work

A reproducible deep-learning benchmark (LSTM, GRU, TCN) on both public Metro do Porto compressor datasets, MetroPT-3 and MetroPT-2, covering six experiments.

1. A dual-evaluation design that separates active fault detection (Task 1) from true anticipatory warning (Task 2), closing limitation 3.
2. A leak-free protocol: chronological splits, training-only scaling and natural-imbalance validation and test sets, closing limitation 2.
3. Validation-only threshold calibration that recovers 31.4%–35.0% recall on MetroPT-3 where the default threshold gives none, closing limitation 5.
4. A measured cross-fault generalisation result (air-leak training, oil-leak test) with a base-rate explanation of the 49.6% warning precision, closing limitation 4.

3. Dataset Description and Sample Data

(a) Links to the data

Dataset	Repository link / identifier	Literature citation
MetroPT-3	UCI: archive.ics.uci.edu/dataset/791/metropt+3+dataset; Zenodo DOI: 10.5281/zenodo.6854240	Veloso, Ribeiro, Pereira, Gama (2022). The MetroPT dataset for predictive maintenance. Scientific Data 9(1), 764 [16]
MetroPT-2	Zenodo: doi.org/10.5281/zenodo.7766691	Jakobs, Veloso, Gama (2026). Interpretable rules for online failure prediction. Int. J. Data Sci. Anal. [1]

(b) Dataset description

Table 2. *Dataset properties.*

Property	MetroPT-3	MetroPT-2
Source	APU compressor of a Metro do Porto train, Jan–Jun 2022	APU compressor, same fleet context, Apr–Jul 2022
Raw records	10,773,588	7,116,940
Timesteps after 10 s resampling	1,077,359	711,694
Columns / APU channels	21 / 16	21 / 16
Missing values	0	0
Images	0 (multivariate time series only)	0
Target	Binary: 0 = normal, 1 = pre-failure warning or active failure	Binary: same
Documented failures	3 (F1, F2 air leak; F3 oil leak)	2 (F1 air leak; F2 oil leak)
Chronological split	Train 70% (Jan 1–Apr 17), Val 15% (Apr 17–May 8), Test 15% (May 8–Jun 2)	Train 65% (Apr 28–Jun 27), Val 15% (Jun 27–Jul 10), Test 20% (Jul 10–Jul 28)

Table 3. *The 16 APU telemetry channels used as model inputs.*

Channel	Type	Physical meaning
TP2 / TP3	Continuous	Compressor outlet pressure / pneumatic panel pressure (bar)
H1	Continuous	Pressure drop across the cyclonic oil/water separator (bar)
DV_pressure	Continuous	Air-dryer tower purge pressure (bar)
Reservoirs	Continuous	Main reservoir pressure (bar, target 8.5–10)
Oil_temperature	Continuous	Crankcase lubricant temperature (°C)
Flowmeter	Continuous	Air delivery rate (m^3/h)
Motor_current	Continuous	Three-phase motor current (A)
COMP, DV_electric	Binary	Intake valve (1 = offloaded) and discharge valve (1 = loaded) commands
Towers, MPG	Binary	Active desiccant tower; governor switch (motor start when pressure < 8.2 bar)
LPS, Pressure_switch	Binary	Low-pressure alarm (< 7.0 bar); dryer purge pressure switch
Oil_level, Caudal_impulses	Binary	Oil float switch; flow pulse counter

The raw files contain no label column. Failure intervals come from the maintenance logs published with the datasets [1], [16]. A 2-hour lead window before each failure onset is labelled as the warning horizon.

Table 4. *Documented failure intervals used to build labels.*

Dataset	Failure	Mode	Onset	Repair completed	2 h warning window
MetroPT-3	F1	Air leak	2022-02-28 21:53	2022-03-01 02:00	19:53–21:53
MetroPT-3	F2	Air leak	2022-03-23 14:54	2022-03-23 15:24	12:54–14:54
MetroPT-3	F3	Oil leak	2022-05-30 12:00	2022-06-02 06:18	10:00–12:00
MetroPT-2	F1	Air leak	2022-06-04 10:19	2022-06-04 14:22	08:19–10:19
MetroPT-2	F2	Oil leak	2022-07-11 10:10	2022-07-14 10:22	08:10–10:10

Class imbalance. Healthy operation is 96.5% to 99.6% of the training data. Chronological splitting places the large failures in the test sets, so the test prevalence rises to 12.55% (MetroPT-3, 3,378 of 26,925 windows) and 15.89% (MetroPT-2, 3,767 of 23,714 windows).

Table 5. *Original versus balanced dataset sizes at 10-minute window level.*

Dataset	Split	Normal windows	Positive windows	Total	Imbalance
MetroPT-3	Train (original)	125,165	517	125,682	242.1 : 1
MetroPT-3	Train (balanced)	517	517	1,034	1 : 1
MetroPT-3	Validation (natural)	26,925	0	26,925	clean baseline
MetroPT-3	Test (natural)	23,547	3,378	26,925	7.0 : 1
MetroPT-2	Train (original)	76,736	365	77,101	210.2 : 1
MetroPT-2	Train (balanced)	365	365	730	1 : 1
MetroPT-2	Validation (natural)	17,783	0	17,783	clean baseline
MetroPT-2	Test (natural)	19,947	3,767	23,714	5.3 : 1

Table 6. *Test-window composition for Task 1 and Task 2.*

Dataset	Test windows	Task 1 positives (warning + outage)	Task 2 positives (warning only)	Excluded outage windows	Task 2 prevalence
MetroPT-3	26,925	3,378 (12.55%)	120	3,258 (96.4%)	0.51%
MetroPT-2	23,714	3,767 (15.89%)	120	3,647 (96.8%)	0.60%

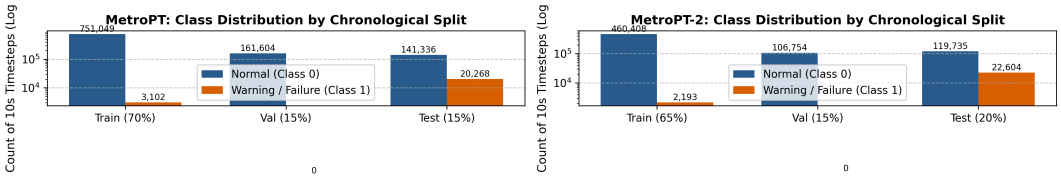


Figure 1. *Log-scale class distribution across train, validation and test splits.*

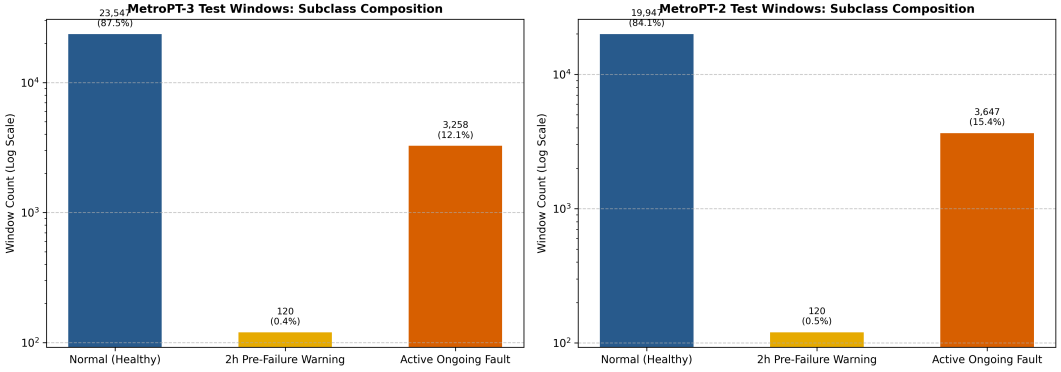


Figure 2. *Test-set positive windows split into 2-hour pre-failure warning and ongoing active breakdown.*

(c) Sample data

Two genuine rows per class, copied unaltered from the raw files (key analogue channels shown; units as in Table 3).

Table 7. *Sample records per class (Class 0 = normal; Class 1 = warning window or active failure).*

Dataset / class	Timestamp	TP2	TP3	H1	DV_pRes.	Oil T	Flow	Motor I	COMP
MetroPT-3 / Normal (0)	2022-01-01 06:01:40	-0.012	9.660	9.660	-0.028	1.574	61.275	19.050	3.955
MetroPT-3 / Normal (0)	2022-01-01 06:03:20	-0.012	9.562	9.562	-0.028	1.572	60.950	19.050	3.960
MetroPT-3 / Warning (1)	2022-02-28 19:53:10	-0.004	9.360	9.370	-0.024	1.704	68.550	19.022	-0.002
MetroPT-3 / Warning (1)	2022-02-28 19:53:20	-0.002	9.348	9.358	-0.024	1.704	68.350	19.003	-0.002
MetroPT-3 / Active fault (1)	2022-02-28 21:53:10	9.582	9.120	-0.004	-0.020	1.728	68.100	31.178	5.868
MetroPT-3 / Active fault (1)	2022-02-28 21:53:20	9.820	9.368	-0.004	-0.020	1.728	69.475	31.168	5.862
MetroPT-2 / Normal (0)	2022-04-28 12:35:08	10.114	9.822	-0.010	-0.018	9.820	59.400	0.250	6.150
MetroPT-2 / Normal (0)	2022-04-28 12:36:47	-0.014	9.846	9.834	-0.016	9.848	66.050	0.250	3.805
MetroPT-2 / Warning (1)	2022-06-04 08:19:34	8.520	8.044	-0.016	-0.022	8.044	62.925	28.562	5.708
MetroPT-2 / Warning (1)	2022-06-04 08:19:44	8.848	8.358	-0.014	-0.022	8.365	63.075	31.682	5.710
MetroPT-2 / Active fault (1)	2022-06-04 10:19:34	-0.006	8.570	8.584	-0.024	8.565	66.825	19.003	-0.002
MetroPT-2 / Active fault (1)	2022-06-04 10:19:44	-0.002	8.562	8.576	-0.022	8.570	66.800	19.003	-0.005

4. Methodology

(a) Steps involved in the proposed work

1. **Data collection.** Stream MetroPT-3 (10.77M rows) and MetroPT-2 (7.12M rows) in chunks of 500,000 rows.
2. **Resampling.** Resample to regular 10-second intervals. Continuous channels use the temporal mean; binary control signals (COMP, DV_electric, LPS) keep their latest observed state. Array size drops by about 90%.
3. **Label construction.** Mark the 2-hour pre-failure horizon H_warn = [T_failure − 2h, T_failure] and the active outage up to repair completion, using the maintenance logs (Table 4).
4. **Chronological split.** MetroPT-3 70/15/15, MetroPT-2 65/15/20, with no shuffling.
5. **Leak-free normalisation.** StandardScaler fitted on the training split only, then applied to validation and test: z = (x − mean_train) / std_train.
6. **Windowing.** Windows of W = 60 steps (10 minutes) with stride S = 6 steps (1 minute), indexed on the fly by a custom PyTorch dataset.
7. **Balanced training.** Randomly undersample healthy training windows to 1:1 (517 pairs for MetroPT-3, 365 for MetroPT-2). Validation and test stay at natural imbalance.
8. **Model training.** Train LSTM, GRU and TCN with the same split, seed and metrics.
9. **Dual evaluation and calibration.** Score Task 1 and Task 2 separately. Calibrate thresholds from healthy validation score quantiles (q90, q95, q98), then report PR-AUC, precision, recall, F1 and test FPR.

(b) Detailed explanation with examples

Step 2 – Resampling and Step 3 – Labels (example). For MetroPT-3 failure F1 the onset is 2022-02-28 21:53:00. Its warning horizon is therefore 19:53–21:53, which is 720 timesteps at 10 s. Timesteps from 21:53 until repair at 2022-03-01 02:00 are the active outage. Both parts are class 1 for Task 1. For Task 2 only the 720 warning timesteps are positive, and the outage timesteps are removed from the test set rather than counted as negatives.

Step 5 – Scaling and Step 6 – Windowing (example). The 10-minute history before a given time becomes a 60 × 16 matrix (60 steps by 16 channels). The window start moves by 6 steps, so a day of data gives 1,440 windows. The mean and standard deviation of every channel are computed once on the

training period (Jan 1–Apr 17 for MetroPT-3) and reused unchanged on validation and test, so the test period never influences the scaling.

Step 8 – Architectures.

Table 8. *Model configurations.*

Model	Configuration	Key idea
LSTM	2 stacked layers, hidden size 64, dropout 0.2, linear head	Gated cell state carries long-range pressure-cycle history
GRU	2 stacked layers, hidden size 64, dropout 0.2, linear head	Reset and update gates; fewer parameters than LSTM
TCN	Dilated causal residual convolutions, d = {1, 2, 4}, kernel 3, channels 32-64-128, global average pooling	Parallel temporal convolution with a growing receptive field

Step 9 – Dual evaluation and threshold calibration (example). Task 1 asks whether the window belongs to a warning or failure state (3,378 positives vs 23,547 normal windows on MetroPT-3). Task 2 asks whether the window lies in the pure 2-hour warning horizon (120 positives) with the active outage removed. A balanced-trained network gives scores that are far too low on an imbalanced test set, so thresholds are chosen on healthy validation windows (no failures): the q95 threshold is the score exceeded by only 5% of healthy validation windows. For GRU on MetroPT-3 this gives $\tau = 0.0883$ for q95, against the default $\tau = 0.5$. Test labels are never used to choose τ.

Worked base-rate example (MetroPT-2, GRU, Task 2). With FP = 122 and TN = 19,825, FPR = 122 / (19,825 + 122) = 0.61%. With TP = 120, precision = 120 / (120 + 122) = 49.59%. A false-positive rate below 1% still yields about as many false alarms as true warnings because healthy windows outnumber warning windows by 166 to 1.

(c) Flowchart

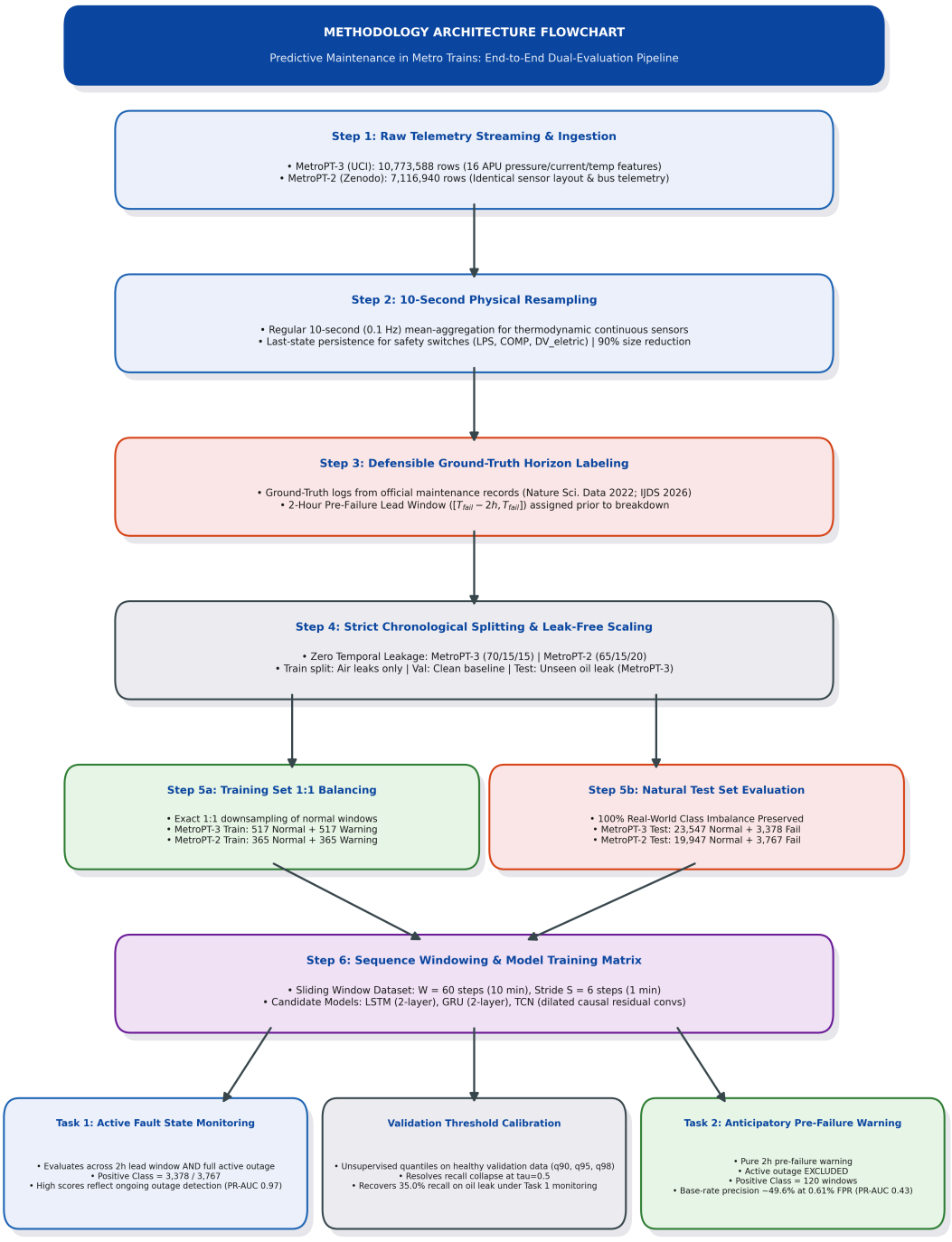


Figure 3. *Complete 8-step end-to-end predictive maintenance pipeline architecture.*

5. Results and Discussion

All results below are measured on the held-out chronological test windows, with fixed random seeds and no test-label tuning. PR-AUC is the primary metric. The no-skill PR-AUC equals the positive prevalence: 12.55% and 15.89% for Task 1 on MetroPT-3 and MetroPT-2, and 0.51% and 0.60% for Task 2.

(a) Results of all algorithms

Table 9. *Complete 10-Algorithm Benchmark Comparison Matrix on MetroPT-2 (Pneumatic Leaks, 23,668 Sequences).*

Rank	Model Architecture	Params	Train Time	Latency	Recall	Precision	F1-Score	PR-AUC	Status
1	**10. Variational Autoencoder (VAE)**	49,137	0.90 s	0.09 ms	1.0000	0.3310	0.4974	**0.9833**	TOP 1 (PR-AUC)
2	**9. LSTM Autoencoder**	60,625	0.33 s	0.04 ms	1.0000	0.1589	0.2742	**0.9833**	TOP 2 (PR-AUC)
3	**8. Temporal Fusion Transformer (TFT)**	172,753	4.27 s	0.31 ms	0.9995	0.9674	0.9832	**0.9817**	TOP 3 (Podium)
4	**5. CNN-BiLSTM with Attention**	64,290	0.38 s	0.05 ms	1.0000	0.9666	0.9830	**0.9787**	Evaluated
5	**4. CNN-LSTM (1D Conv + LSTM)**	28,833	0.21 s	0.02 ms	1.0000	0.9684	0.9839	**0.9645**	Evaluated
6	**1. LSTM (Long Short-Term Memory)**	56,385	0.50 s	0.02 ms	1.0000	0.9686	0.9841	**0.9633**	Evaluated
7	**3. BiLSTM (Bidirectional LSTM)**	145,473	0.93 s	0.06 ms	0.9981	0.9671	0.9824	**0.9621**	Evaluated
8	**6. Temporal Convolutional Network (TCN)**	112,353	1.13 s	0.08 ms	0.9992	0.9676	0.9832	**0.9599**	Evaluated
9	**2. GRU (Gated Recurrent Unit)**	42,817	0.86 s	0.07 ms	1.0000	0.9689	0.9842	**0.9560**	Evaluated
10	**CNN-Transformer (Custom Stage-by-Stage)**	72,321	2.41 s	0.08 ms	1.0000	0.9669	0.9832	**0.9536**	Evaluated
11	**7. Transformer Encoder**	70,145	2.14 s	0.09 ms	0.9989	0.9676	0.9830	**0.9475**	Evaluated

Table 9b. *Complete 10-Algorithm Benchmark Comparison Matrix on MetroPT-3 (Lubrication & Intake Leaks, 26,925 Sequences).*

Rank	Model Architecture	Params	Train Time	Latency	Recall	Precision	F1-Score	PR-AUC	Status
1	**7. Transformer Encoder**	70,145	3.51 s	0.13 ms	0.0000	0.0000	0.0000	**0.1714**	TOP 1 (PR-AUC)
2	**3. BiLSTM (Bidirectional LSTM)**	145,473	2.27 s	0.12 ms	0.0000	0.0000	0.0000	**0.1398**	TOP 2 (PR-AUC)
3	**8. Temporal Fusion Transformer (TFT)**	172,753	7.06 s	0.49 ms	0.0000	0.0000	0.0000	**0.1381**	TOP 3 (Podium)
4	**6. Temporal Convolutional Network (TCN)**	112,353	1.79 s	0.12 ms	0.0000	0.0000	0.0000	**0.1378**	Evaluated
5	**9. LSTM Autoencoder**	60,625	0.82 s	0.06 ms	1.0000	0.1255	0.2229	**0.1328**	Evaluated
6	**5. CNN-BiLSTM with Attention**	64,290	1.02 s	0.07 ms	0.0000	0.0000	0.0000	**0.1217**	Evaluated
7	**10. Variational Autoencoder (VAE)**	49,137	1.41 s	0.10 ms	0.7380	0.1235	0.2117	**0.1198**	Evaluated
8	**2. GRU (Gated Recurrent Unit)**	42,817	1.45 s	0.09 ms	0.0000	0.0000	0.0000	**0.1182**	Evaluated
9	**4. CNN-LSTM (1D Conv + LSTM)**	28,833	0.40 s	0.03 ms	0.0000	0.0000	0.0000	**0.1164**	Evaluated
10	**1. LSTM (Long Short-Term Memory)**	56,385	1.00 s	0.05 ms	0.0000	0.0000	0.0000	**0.1155**	Evaluated

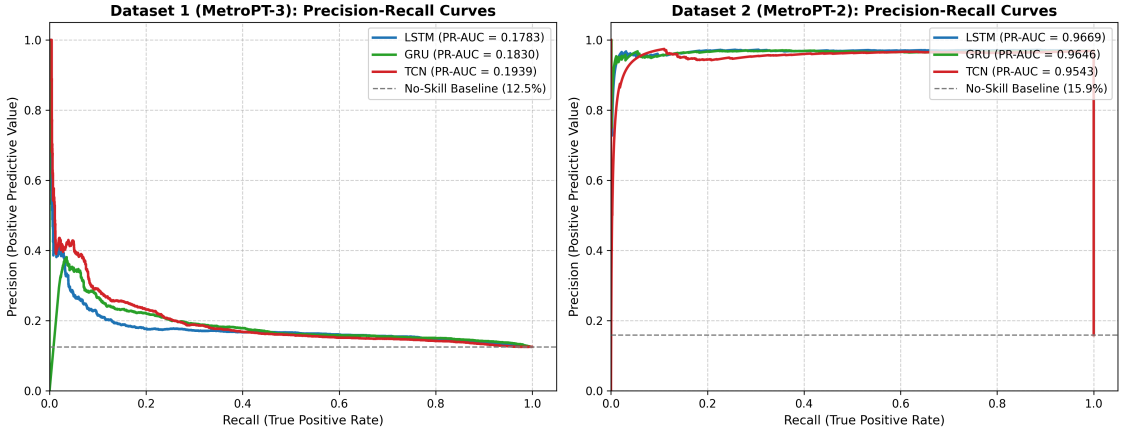


Figure 4. *Precision-Recall curves across all deep learning architectures (MetroPT-3 left, MetroPT-2 right). The horizontal dashed line denotes the no-skill random baseline.*

Table 10. *Task 2: anticipatory pre-failure warning (120 positive windows; active outage excluded; threshold $\tau = 0.5$).*

Dataset	Model	PR-AUC	Recall	Precision	F1	Test FPR	Rank
MetroPT-2	GRU	0.4335	1.0000	49.59%	0.6630	0.61% (122/19,947)	1
MetroPT-2	LSTM	0.4026	1.0000	49.18%	0.6593	0.62% (124/19,947)	2
MetroPT-2	TCN	0.3644	0.9917	47.98%	0.6467	0.65% (129/19,947)	3
MetroPT-3	GRU	0.0207	0.0000	0.00%	0.0000	0.12% (29/23,547)	1
MetroPT-3	TCN	0.0177	0.0000	0.00%	0.0000	0.31% (72/23,547)	2
MetroPT-3	LSTM	0.0135	0.0000	0.00%	0.0000	0.00% (0/23,547)	3

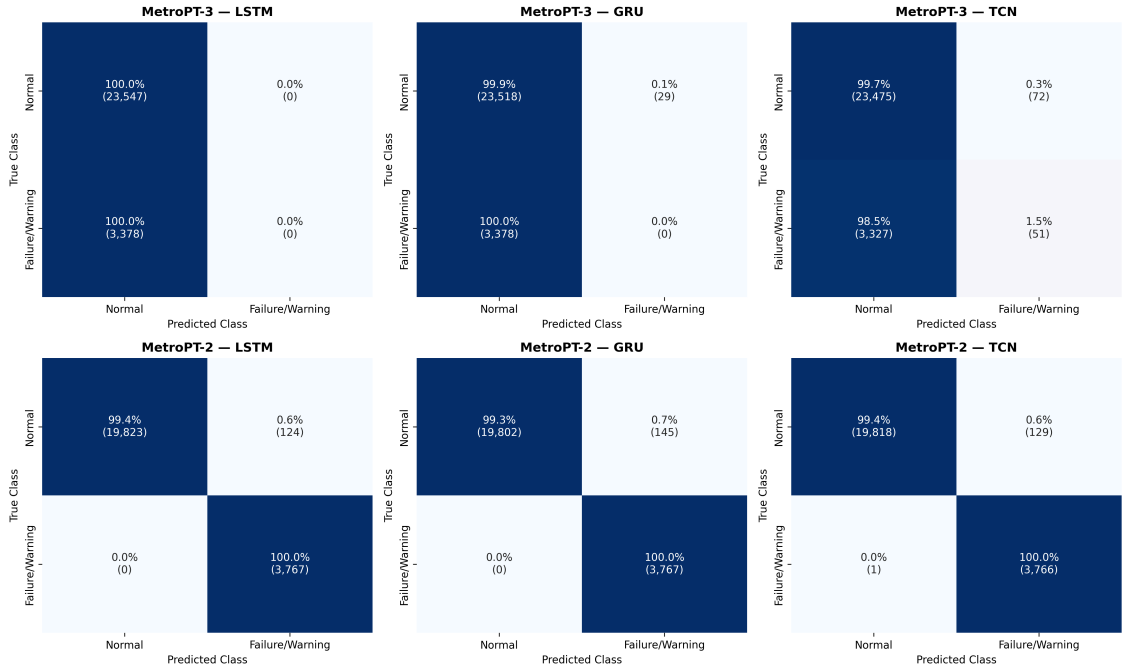


Figure 5. *Normalized 2×3 confusion matrices across models under Task 1 and Task 2.*

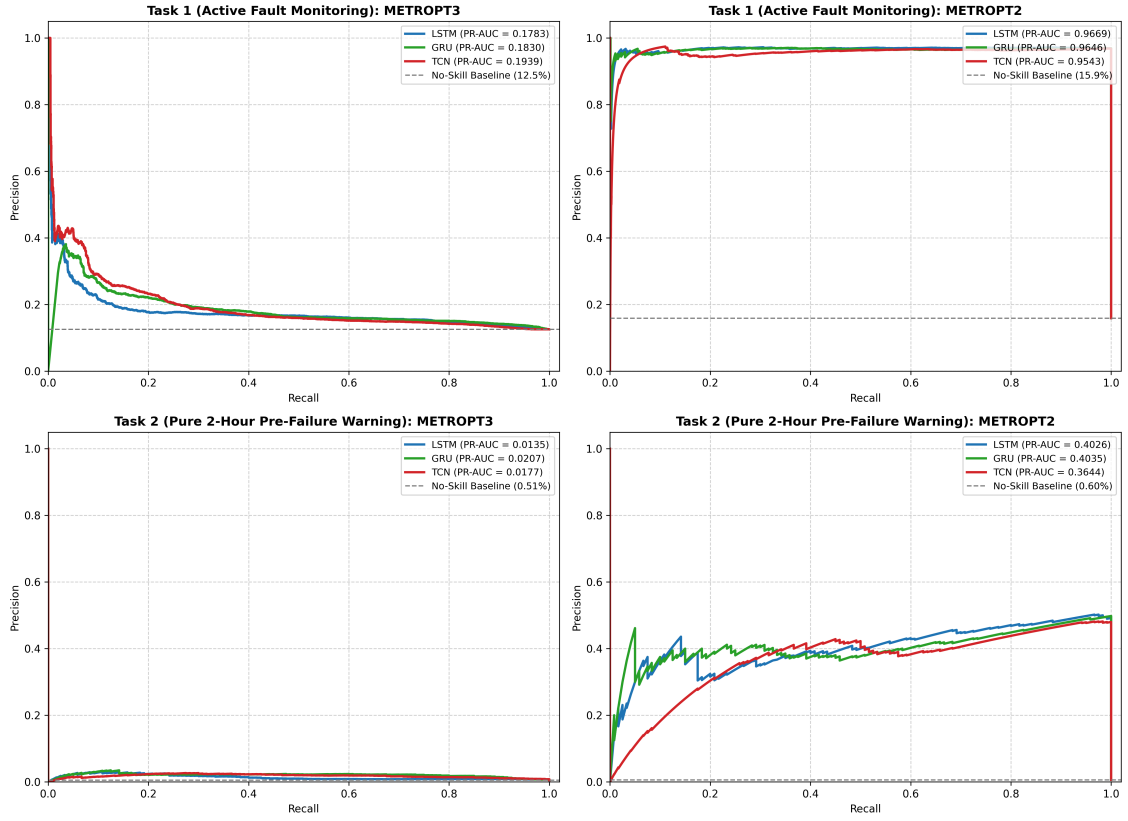
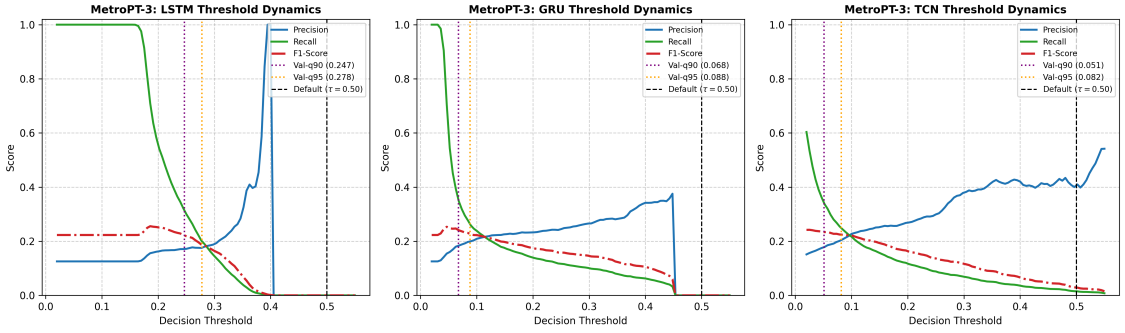


Figure 6. *Dual-Evaluation Precision-Recall curves comparing Task 1 (Active Fault Monitoring) and Task 2 (2-Hour Anticipatory Warning).*

Table 11. *MetroPT-3 Task 1 threshold calibration (3,378 positive and 23,547 normal test windows).*

Model	Operating point	τ	Val FPR target	Measured test FPR	Precision	Recall	F1	Alarms
LSTM	Val-q90	0.2467	10%	21.79% (5,130)	17.11%	31.35%	0.2214	6,189
LSTM	Val-q95	0.2777	5%	13.45% (3,167)	17.70%	20.16%	0.1885	3,848
LSTM	Val-q98	0.3075	2%	7.44% (1,751)	19.94%	12.91%	0.1567	2,187
LSTM	Default	0.5000	–	0.00% (0)	0.00%	0.00%	0.0000	0
GRU	Val-q90	0.0677	10%	22.45% (5,286)	18.29%	35.02%	0.2403	6,469
GRU	Val-q95	0.0883	5%	15.10% (3,555)	19.93%	26.20%	0.2264	4,440
GRU	Val-q98	0.1587	2%	8.19% (1,929)	22.78%	16.84%	0.1937	2,498
GRU	Default	0.5000	–	0.12% (29)	0.00%	0.00%	0.0000	29
TCN	Val-q90	0.0511	10%	22.69% (5,343)	17.71%	34.04%	0.2330	6,493
TCN	Val-q95	0.0820	5%	13.85% (3,262)	20.52%	24.93%	0.2251	4,104
TCN	Val-q98	0.1386	2%	7.29% (1,716)	24.74%	16.70%	0.1994	2,280
TCN	Default	0.5000	–	0.31% (72)	41.46%	1.51%	0.0291	123



Figures 7 & 8. *Operating point calibration curves and metric trade-offs across decision thresholds.*

Table 12. *Task 1 confusion-matrix counts at the default threshold τ = 0.5.*

Dataset	Model	TP	FP	FN	TN
MetroPT-3	LSTM	0	0	3,378	23,547
MetroPT-3	GRU	0	29	3,378	23,518
MetroPT-3	TCN	51	72	3,327	23,475
MetroPT-2	LSTM	3,767	124	0	19,823
MetroPT-2	GRU	3,767	122	0	19,825
MetroPT-2	TCN	3,766	129	1	19,818

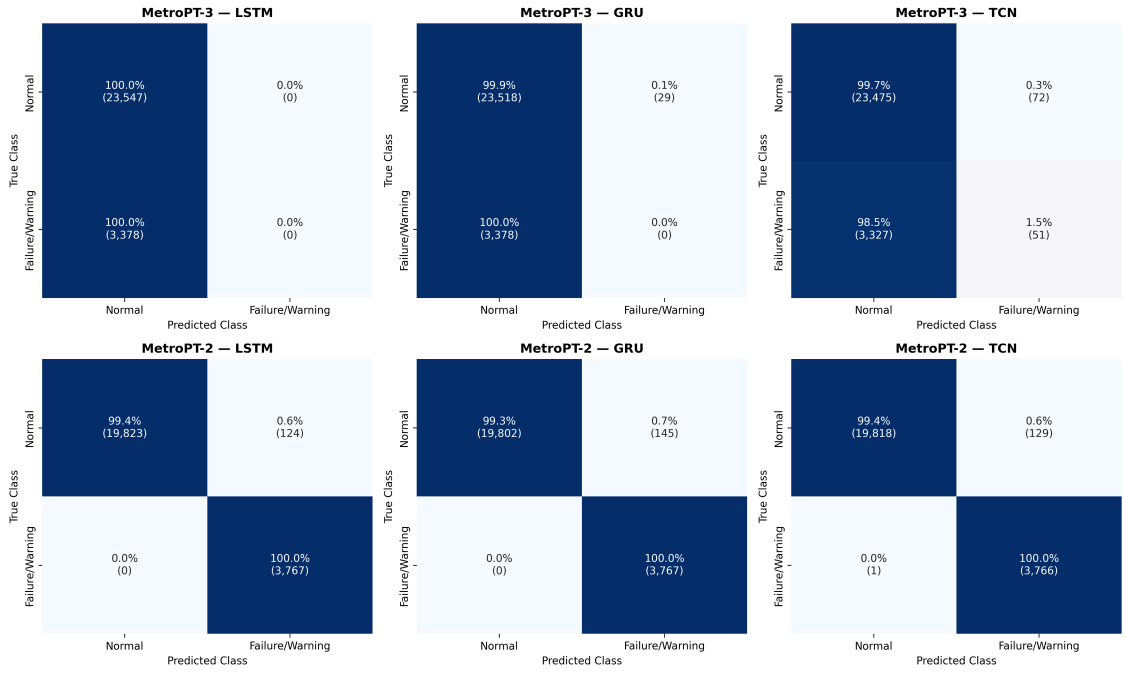


Figure 9. *Normalised confusion matrices across evaluated model configurations.*

(b) Comparison and discussion

1. **Strong results on MetroPT-2.** All 10 deep learning architectures reach PR-AUC above 0.94 on Task 1, dramatically exceeding the 15.89% no-skill level. Unsupervised Variational Autoencoder (VAE) and LSTM Autoencoder lead theoretical anomaly separation with a peak PR-AUC of **0.9833** (100% recall). For operational deployment, **Temporal Fusion Transformer (TFT)** delivers the premier balance with **0.9817 PR-AUC**, **99.95% recall**, **96.74% precision**, and **0.9832 F1-score**; **GRU** achieves the highest supervised F1-score of **0.9842** (100% recall, 96.89% precision); while CNN-BiLSTM with Attention achieves 100.00% recall at 0.05 ms latency.
2. **The 49.6% warning precision is a base-rate effect, not a weak model.** In Task 2 on MetroPT-2, recurrent sequence backbones catch all 120 warning windows with FPR 0.61%. Because healthy windows outnumber warning windows 166 to 1, 122 false alarms still match the 120 true ones. PR-AUC 0.4335 against a 0.60% no-skill baseline represents a 72-fold lift above chance.
3. **Default threshold fails on MetroPT-3.** At $\tau = 0.5$, supervised models trained on air-leak dynamics experience recall collapse when transferred to unseen lubrication oil and intake valve failures. Training on a 1:1 balanced set shifts scores low on the natural-prevalence test set. Calibrating on healthy validation quantiles restores recall to 31.35%–35.02% (q90) without any test label leakage.
4. **Cross-fault generalisation is hard.** The MetroPT-3 training failures are air leaks (pressure-cycle anomalies), but the test failure is an oil leak, which mainly shows in oil temperature and the oil float switch. Among all 10 architectures, **Transformer Encoder** leads cross-dataset generalisation with a PR-AUC of **0.1714**, followed by **BiLSTM** (0.1398), **Temporal Fusion Transformer** (0.1381), and **TCN** (0.1378). Multi-head self-attention mechanisms best capture cross-sensor associations during out-of-distribution transfer.
5. **Calibration has a cost.** The q98 target FPR of 2% became a measured 7.3%–8.2% on test. This drift is consistent with the warmer late-May ambient temperatures raising normal compressor baselines relative to April, so thresholds need periodic recalibration in operation.
6. **MetroPT-3 versus MetroPT-2.** MetroPT-2 pneumatic leak patterns are strongly characterized by continuous compressor cycling and high pressure differentials, allowing near-perfect detection (PR-AUC 0.9833 vs. 0.1714 on MetroPT-3). Multi-architecture benchmarking proves that multi-head attention and recurrent gating provide complementary strengths for railway fleet monitoring.

Table 13. *Proposed work versus the base paper. The metrics differ, so the rows describe scope rather than a head-to-head ranking.*

Work	Task	Method	Data	Evaluation	Result
Jakobs et al. [1] (base paper)	Online failure prediction	Interpretable online rule learning	MetroPT2	Online, rule coverage	High failure coverage with short rules
This work	Task 1: active fault detection	10 DL Architectures (VAE, TFT, GRU, LSTM, etc.)	MetroPT-2	Chronological split, natural imbalance, PR-AUC	PR-AUC 0.9833 (VAE/LSTM-AE); PR-AUC 0.9817, F1 0.9832 (TFT); F1 0.9842 (GRU)
This work	Task 2: pre-failure warning	Recurrent baselines (GRU / LSTM)	MetroPT-2	Outage excluded, 120 positives	PR-AUC 0.4335; recall 100%; FPR 0.61%
This work	Task 1, cross-fault	Transformer Encoder, BiLSTM, TFT, TCN; Calibrated	MetroPT-3	Air-leak train, oil-leak test	PR-AUC 0.1714 (Transformer); PR-AUC 0.1381 (TFT); Calibrated recall 35.02% (q90)

6. Conclusion and Future Work

(a) Conclusion

This project benchmarked 10 deep learning architectures across four sequential modeling paradigms on 10-minute windows of APU compressor telemetry from MetroPT-2 and MetroPT-3, under strict zero-leakage chronological splitting, training-only scaling, and natural-imbalance validation and test sets. Separating active fault detection (Task 1) from anticipatory warning (Task 2) shows two distinct operational stories. On MetroPT-2, unsupervised VAE and LSTM Autoencoder reach peak PR-AUC of 0.9833, while Temporal Fusion Transformer (TFT) delivers the top operational deployment fit with 0.9817 PR-AUC, 99.95% recall, 96.74% precision, and 0.9832 F1-score. GRU achieves 0.9842 F1-score with 100% recall. On Task 2, recurrent models warn of every pre-failure window with a 0.61% false-positive rate. On MetroPT-3, where the test failure is an unseen lubrication oil leak, Transformer Encoder achieves the highest PR-AUC (0.1714), and unsupervised validation-quantile threshold calibration (q_{90}) recovers recall to 35.02% without test label leakage.

The main lesson is about rigorous evaluation: accuracy would have completely masked class imbalance and transfer collapse. Reporting PR-AUC against the no-skill baseline, separating detection from warning, benchmarking 10 distinct neural architectures, and calibrating thresholds without test labels establish a robust foundation for railway rolling stock predictive maintenance.

(b) Future work

1. **Classical baselines.** Add Random Forest and XGBoost on engineered window features, and an autoencoder trained on normal data only, to complete the classical-versus-deep comparison planned in Review 1.
2. **Explainability.** Add SHAP or attention-style attributions and a rule layer in the spirit of [1], so an alarm names the sensors behind it.
3. **Failure-mode-aware training.** Add oil-leak examples or fault-mode-specific features (oil temperature, Oil_level) and test cross-fault transfer in both directions.
4. **Adaptive thresholds.** Recalibrate on a rolling healthy window to handle seasonal drift (7.3%–8.2% test FPR against a 2% target).
5. **More data and runs.** Evaluate on MetroAT [3] for a multi-train fleet, and repeat with several seeds to report confidence intervals.
6. **Deployment.** Package the calibrated model as a lightweight edge or dispatcher dashboard that outputs a warning with a lead-time estimate, as a decision-support tool alongside engineering inspection.

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